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BUILD WITH BHARAT 4.0

NATIONAL LEVEL HACKATHON 2026

PROJECT NAME	CrediFlow
TAGLINE	AI-Assisted GST Reconciliation & Vendor Resolution Platform for MSMEs
PROBLEM STATEMENT	Automated GST ITC Reconciliation & Closed-Loop Vendor Dispute Resolution under Rule 60 CGST
HACKATHON TRACK	FinTech & B2B SaaS Workflow Automation
TEAM NAME	Team Nexora
TEAM MEMBERS	• Anand Awasthi — Full-Stack & System Architecture • Vaishnavi Dwivedi — Backend & APIs • Harleen Kaur — Research & Presentation
COLLEGE / INSTITUTION	KCC Institute of Technology and Management, Greater Noida

| PROBLEM STATEMENT

The Working Capital Trap: Section 16(2)(aa) & Broken Vendor Reconciliation

The Core Statutory Bottleneck: Under **Section 16(2)(aa)** of the CGST Act and **Rule 60**, a business can claim Input Tax Credit (ITC) *only* when the supplier correctly uploads invoices to GSTR-1 and it reflects in the buyer's **GSTR-2B**. Any error (wrong GSTIN, mismatched value, rate discrepancy, or missing filing) locks ITC instantly. The buyer is forced to pay that tax in cash out of pocket while chasing vendors manually.

The Real-World Problem Cascade 1. Invoice Raised: Vendor issues invoice; goods received & payment made. 2. Data Mismatch: Purchase Register != GSTR-2B (vendor error/omission). 3. Locked ITC: Working capital frozen; cannot claim credit on portal. 4. Manual Excel Audit: Finance teams burn 30+ hours/month manually checking. 5. Friction-Heavy Chasing: Unstructured WhatsApp calls, emails, generic notices. 6. Compliance Exposure: Risk of departmental DRC-01C notices + 18% interest.	Target Users & Market Impact • Primary: MSME finance & accounts teams managing 30+ recurring vendors. • Secondary: Chartered Accountant (CA) firms managing multi-client portfolios. • Scale of Impact: Average MSME loses 3-7% of monthly turnover to locked ITC. Core Insight: <i>"Finding the mismatch is easy. Closing it is the real problem."</i>
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| PROPOSED SOLUTION

CrediFlow: Closed-Loop GST Resolution Pipeline vs Passive Reporting

The Market Gap: Existing tools (ClearTax, Zoho Books, Tally) only collect data and output a passive Excel discrepancy report. **Detection without resolution leaves 100% of the chasing burden on the buyer.** CrediFlow transforms passive mismatch detection into an autonomous, closed-loop resolution workflow.

Stage	What CrediFlow Does	Core Value to Business
1. DETECT	Deterministically matches Purchase Register with GSTR-2B using exact & near-match rules.	Sub-second ingestion; catches every typo, date, and rate error.
2. QUANTIFY	Calculates exact blocked ITC per invoice, per vendor, and by tax head (CGST/SGST/IGST).	Clear visibility into exact working capital at risk.
3. NUDGE	Generates professional bilingual (English + Hindi) follow-ups with Section 16(2)(aa) legal citations.	Zero-friction communication vendors actually understand and act on.
4. RESOLVE	Dual-agent AI identifies statutory root cause and gives vendors step-by-step amendment guidance.	Eliminates back-and-forth confusion; guides GSTR-1 amendment.
5. VERIFY	Re-reconciles after vendor filing; marks resolved only on confirmed GSTR-2B match.	Closed-loop guarantee; unlocks cash credit safely.

Core Architectural Philosophy: **Deterministic Engine = Truth** (100% rule-based math; zero LLM hallucinations in tax calculations) | **AI = Intelligence Around Truth** (Classification, plain-language guidance, and bilingual communication).

FLOW OF SOLUTION

End-to-End Autonomous Workflow & System Architecture

End-to-End Closed-Loop Workflow Ingestion: Multi-format upload of Purchase Register (PR) & GSTR-2B (Excel/CSV/JSON). Normalization & Matching: Deterministic matching hierarchy (Exact Match ± Rs. 2 -> Levenshtein <= 2 -> Tax-Head Error -> Missing in 2B). Exposure Quantification: Computes exact INR exposure per invoice & supplier leaderboard. Dual-Agent AI Audit: Agent A (Root-Cause Classifier) + Agent B (Audit Cross-Examiner). Human-in-the-Loop Gate: Auto-routes exposure > Rs. 50,000 or confidence < 85% for review. Dispatch: 1-Click WhatsApp deep-link (wa.me) / Email + Court-Ready ReportLab PDF notice. Status Tracking: Detected -> Nudged -> Pending -> Verified. Closed-Loop Verification: Re-runs matching upon vendor filing; marks resolved.	Three Supporting Core Engines Risk Quantification Engine: Calculates exact rupee exposure per tax head (CGST, SGST, IGST, Cess). 100% deterministic arithmetic. Vernacular Message Generator: Generates culturally nuanced, professional English + Hindi follow-up messages tailored to vendor risk score. Closed-Loop Verification Engine: Automatically re-reconciles after vendor amendment filings to guarantee that credit is safe before closing tickets. Safety & Compliance Guardrail: Strict statutory grounding under Section 16(2)(aa) & Rule 60. Financial managers retain full override control at all times.
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TECH STACK

Production Architecture & Implementation Approach

Layer	Core Technologies	Engineering Implementation & Role
Frontend UI	React 19.2, Vite 8.2, Tailwind CSS, Lucide React, Canvas Confetti	Single-page reactive dashboard, live audit ledger, HITL review modal, interactive charts.
Backend API	Python 3.10+, FastAPI 0.115+, Uvicorn, Pydantic v2, HTTPX, OpenPyXL	Asynchronous REST microservices, multi-format Excel/CSV/JSON file parsing, strict validation.
Persistence	SQLite 3 (crediflow.db) + FastAPI In-Memory Session Cache	Persists audit ledgers, benchmark runs, human decisions, and notice dispatches; instant live cache.
Data Pipes	Tinybird / ClickHouse SQL Pipes (.pipe format)	Streaming analytics, high-volume batch ingestion, and aggregate risk reporting.
AI Layer	RocketRide Dual-Agent Pipeline (OpenAI backend) + Deterministic Fallback	Agent A Root Cause + Agent B Cross Examiner + 100% offline statutory fallback.
Doc Engine	ReportLab 5.0.1 + Custom Bilingual Template Engine	Programmatically builds legal Section 16(2)(aa) PDF demand notices & bilingual nudges.
Deployment	Vercel Serverless (Python ASGI runtime + static React assets)	Cloud serverless deployment with fallback disk paths and live production routing.

Core Implementation Principle: *"Deterministic Truth + Generative Agility"* — Every rupee figure, tax rate, and compliance exposure is calculated by pure mathematical rule logic. AI is strictly bounded to statutory reasoning, human-readable explanations, and culturally nuanced communication.

| UNIQUE SELLING PROPOSITION (USP)

Why CrediFlow Outperforms Existing Tax & ERP Tools

Capability / Feature	Typical GST / ERP Tools (ClearTax, Tally, Zoho)	CrediFlow Platform
Mismatch Detection	Basic rule checking	Multi-tier Engine: Exact ($\pm$ Rs. 2) -> Levenshtein ≤ 2 -> Tax Head -> Missing
Exact ITC Quantification	Limited / Aggregate only	Full Granularity: Per invoice + per vendor + by tax head (CGST/SGST/IGST)
Root-Cause Explanation	Rare / Raw error codes	Dual-Agent AI: Statutory root cause + plain-language amendment guidance
Vendor Follow-up	Manual calls / Copy-pasting	1-Click Bilingual Nudges: WhatsApp (wa.me) & Email (English + Hindi)
Legal Notice Generation	None (Manual lawyer draft)	Auto-Generated PDF Notices: Formal Section 16(2)(aa) legal demand notices
Resolution Tracking	None (Software stops at report)	Built-in Lifecycle: Detected -> Nudged -> Pending -> Verified
Post-Correction Verification	None (Manual re-check)	Automated Re-reconciliation: Closes case only upon confirmed GSTR-2B match

Strategic Differentiation: Most tools answer "What is mismatched?". CrediFlow answers: "How do we resolve this and recover the working capital?"

| FEASIBILITY & COMPETITORS

Technical Feasibility, Risk Mitigations & Competitor Matrix

Technical Feasibility & Scale

- **Proven, Robust Stack:** Python FastAPI backend with React 19 frontend guarantees high availability.
- **Standardized Formats:** Works with standard GSTR-2B JSON/Excel portal extracts and ERP books.
- **Sub-Second Throughput:** Tested up to 10,000+ invoices with <2s reconciliation runtime.
- **Zero-Friction Adoption:** No complex API gateway setup required; works out of the box with file uploads.

Competitor Positioning

- **Legacy ERPs (Tally, Zoho):** Focus on accounting entries; zero vendor follow-up workflows.
- **Enterprise Tax Tools (ClearTax):** High cost, enterprise-heavy, passive reports without closed-loop tracking.
- **Generic AI Chatbots:** Unreliable with financial math; hallucinate numbers without deterministic audit trails.

Key Challenges & Engineering Mitigations

- 1. Messy Invoice Formats:**
-> *Mitigation:* Multi-stage string normalization + fuzzy Levenshtein matching + manual override.
- 2. Vendor Non-Response:**
-> *Mitigation:* Prioritized by financial exposure; automated escalation from polite reminder to legal notice.
- 3. Trust in AI & Calculations:**
-> *Mitigation:* AI is never permitted to touch calculations; deterministic engine is the sole source of truth.
- 4. Data Sensitivity & Privacy:**
-> *Mitigation:* Local SQLite persistence + role-based access; zero vendor data shared externally.

| RESEARCH & REFERENCE

Statutory Framework, Live Deployment & Project Links

Statutory & Legal References

- **CGST Act Section 16(2)(aa):** Mandatory condition that ITC can only be availed if invoice details are furnished by the supplier in GSTR-1 and communicated in GSTR-2B.
- **Rule 60 CGST:** Form and manner of furnishing details in GSTR-2A and auto-drafted GSTR-2B statement.
- **Rule 36(4) CGST:** Statutory restriction on ITC claims for un-uploaded invoices.
- **DRC-01C Mechanism:** Automated GST portal intimation of difference in ITC available in GSTR-2B vs claimed in GSTR-3B.

Live Demo, Repository & Deployment

- **Live Web Application:**
<https://credi-flow-gray.vercel.app/>
- **GitHub Repository:**
<https://github.com/Vaishnavi5200/CrediFlow>
- **Interactive Demo Dataset:**
Includes official 45-invoice real-world mismatch dataset + 1,000-invoice scale benchmark.
- **Demo Credentials / Access:** Publicly accessible, zero setup required.

Summary & Vision: CrediFlow bridges the gap between passive tax reports and active cash recovery. By enforcing **DETECT -> QUANTIFY -> NUDGE -> RESOLVE -> VERIFY**, CrediFlow empowers Indian MSMEs to reclaim locked working capital with statutory accuracy and zero friction.